

Hairy cell leukemia

Indolent B-cell leukemia defined by BRAF V600E · splenomegaly with pancytopenia and monocytopenia · one course of a purine analog buys most patients a decade or more; the old “variant” (now SBLPN) is a different, BRAF wild-type disease

Compiled from primary trial reports and consensus guidelines (Hairy Cell Leukemia Foundation consensus, Grever 2017 and 2026 update; NCCN) verified against PubMed, September 2026.

PRESENTATION

- **Rare:** <2% of leukemias; middle-aged adults, strong male predominance
- **Splenomegaly** (often massive) **without adenopathy**, and **pancytopenia with monocytopenia** — the monocytopenia is the clue on a routine CBC
- **Dry tap:** reticulin fibrosis defeats the aspirate — diagnose on the **core biopsy** (interstitial “fried-egg” infiltrate) + **peripheral blood flow**, which finds the few circulating cells
- **Infections** are the cause of morbidity: bacterial, fungal and **atypical mycobacteria** (monocytopenia + neutropenia)
- **Smear:** small-to-medium cells with oval nuclei and **circumferential hairy cytoplasmic projections**

CLASSIC HCL VS ITS LOOK-ALIKES

	Classic HCL	SBLPN (ex-HCL-v)
CD11c · CD103	Bright + · +	+ · +
CD25	+	–
CD123	+ (bright)	usually –
Annexin A1 (IHC)	+ — 100% specific <i>Falini, Lancet 2004</i>	–
TRAP	+ (historical stain)	variable
BRAF V600E	~All (48/48; >95% in 2026 consensus)	Wild-type
Other driver	—	MAP2K1 in ~50% <i>Waterfall 2014</i>
Blood count	Pancytopenia, low monocytes	Leukocytosis, monocytes preserved
Purine analog	Excellent, durable	Poorer, less durable

HCL score (CD11c, CD103, CD123, CD25): 3–4 = classic. WHO-HAEM5 dropped “HCL-variant” and folded it into splenic B-cell lymphoma/leukemia with prominent nucleoli (SBLPN); ICC 2022 keeps HCL-v as a provisional entity.

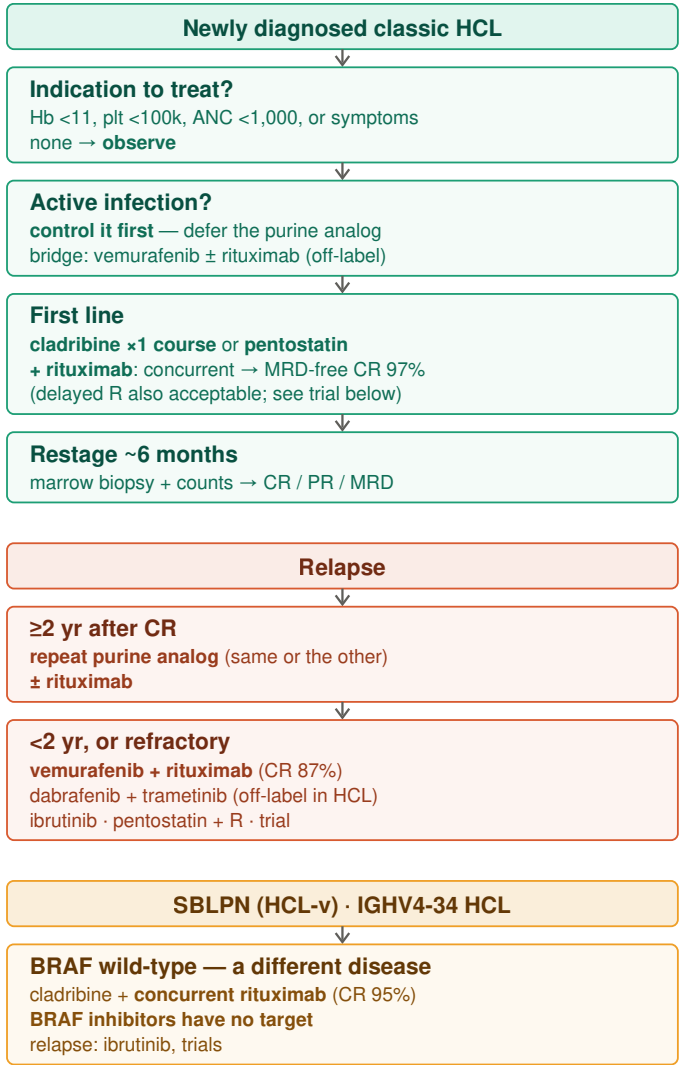
WHEN TO TREAT — CONSENSUS

- **Any of:** Hb <11 g/dL, platelets <100,000/μL, ANC <1,000/μL
- **Symptomatic splenomegaly**, progressive cytopenias, recurrent infection or constitutional symptoms
- **None of these** → **observe**; many patients sit for years
- **CR:** Hb >11 without transfusion, platelets >100,000, ANC >1,500, no hairy cells on marrow or blood, spleen regressed. MRD by IHC or flow is prognostic but not yet a treatment target
- **Relapse cutoff is 2 years** (raised from 1): relapse ≥2 yr after CR → retreat with purine analog; <2 yr → switch class

IGHV4-34 — the trap. Classic-looking HCL that is **BRAF wild-type**: all 13 IGHV4-34+ cases were wild-type *Xi, Blood 2012*. Usually unmutated, higher WBC, **poor response to cladribine** and shorter PFS and OS *Arons, Blood 2009*; MAP2K1 mutations cluster here too. **A BRAF-negative “HCL” should prompt a second look.**

Treat the infection first. Cladribine and pentostatin deepen neutropenia for weeks, and post-cladribine fever is common — so **defer the purine analog during active infection**. If counts are too low for the infection to clear, a **BRAF inhibitor (vemurafenib ± rituximab) restores counts without myelosuppression** and can bridge to definitive therapy — off-label.

TREATMENT PATHWAY



Hairy cell leukemia

KEY EVIDENCE

Study	Result	Read it this way
Scripps cladribine <i>n=358</i>	Single 7-day infusion: CR 91% ; 26% relapsed at a median 29 mo; retreatment CR 62%	The regimen that made HCL a one-course disease
Else, long-term <i>n=233, 16 yr f/u</i>	Pentostatin or cladribine: CR 80%, median RFS 16 yr ; no difference between drugs	Relapse continues late — 25% of 5-yr CRs relapse by 15 yr
Intergroup <i>Grever 1995</i>	Pentostatin vs interferon-α : CR 76% vs 11% . 10-yr OS 81% (Flinn 2000)	Pentostatin is equivalent to cladribine; no excess second cancers
SAKK 32/98 <i>n=100, randomized</i>	Cladribine daily ×5 vs weekly ×5 : response 86% in both	Weekly is not less toxic (grade 3–4 neutropenia 90% vs 80%)
NIH, Chihara <i>n=68, randomized ph2</i>	Cladribine + concurrent vs delayed (≥6 mo) rituximab : CR 100% vs 88% (NS); MRD-free CR 97% vs 24% ; at 8 yr 94% vs 12% still MRD-free	Concurrent R buys deep, lasting MRD negativity at the cost of grade 3–4 thrombocytopenia 59% vs 9% . Survival benefit not shown
MDACC <i>n=139, single-arm</i>	Cladribine → rituximab: CR 97% , 10-yr EFS 87% , OS 91%	Frontline classic HCL relapse only 3.6%
Vemurafenib <i>Tiacci 2015</i>	Relapsed: ORR 96–100% , CR 35–42%; RFS 19 mo in CR, 6 mo in PR	High response, short remission alone
Vem + R <i>Tiacci NEJM 2021, n=30</i>	Median 3 prior lines: CR 87% (65% MRD-negative); PFS 78% at median 37 mo	The preferred non-purine regimen at relapse
Vem + obinutuzumab <i>frontline, n=30</i>	CR 90% , 96% of CRs MRD-negative; no relapse at 35 mo	Chemo-free frontline option; single-arm
Dab + tram <i>ROAR, n=55</i>	Relapsed: ORR 89% , CR 66% ; 24-mo PFS 94%	Not FDA-approved for HCL — the tumor-agnostic label covers solid tumors only
Ibrutinib <i>n=37</i>	Best ORR 54% ; 36-mo PFS 73%	Primary endpoint not met ; modest
Moxetumomab <i>n=80</i>	Anti-CD22 immunotoxin: CR 41%, durable CR 30%; HUS 7.5%, capillary leak 5%	Withdrawn from the US market in 2023 (commercial decision)
SBLPN: clad + R <i>n=20</i>	Concurrent rituximab: CR 95% , 5-yr PFS 63%	TP53-mutant cases fare worse

DOSING

Regimen	Dosing
Cladribine	0.14 mg/kg/day IV over 2 h, days 1–5 (NIH: 0.15 mg/kg), or 0.1 mg/kg/day continuous IV ×7 days , or 0.14 mg/kg weekly ×5 — one course
Clad + rituximab	Cladribine 0.15 mg/kg days 1–5 + rituximab 375 mg/m² weekly ×8 , concurrent from day 1 (or delayed ≥6 mo)
Pentostatin	4 mg/m² IV every 2 weeks until best response; hydrate; hold for renal dysfunction
Vemurafenib + R	Vemurafenib 960 mg PO BID ×8 weeks + rituximab 375 mg/m² ×8 doses over 18 weeks
Dabrafenib + trametinib	Dabrafenib 150 mg BID + trametinib 2 mg daily
Ibrutinib	420 mg daily

MONITORING AND SUPPORTIVE CARE

- **Prolonged CD4 lymphopenia** after purine analogs — median ~**40 months** to recover → **PJP and HSV/VZV prophylaxis**
- **Irradiated blood products indefinitely** after a purine analog (transfusion-associated GVHD)
BSH 2020
- **Neutropenic fever in ~42%** after cladribine. **Routine G-CSF not recommended**: it shortened neutropenia (9 vs 22 days) but did not cut fever or admissions
- **Vemurafenib**: rash, photosensitivity, arthralgia, **cutaneous SCC / keratoacanthoma** — dermatology checks. **Dab/tram**: pyrexia
- Inactivated vaccines before therapy; **no live vaccines**

SPLENIC SMALL B-CELL DIFFERENTIAL

Entity	Distinguishing features
Classic HCL	CD25+ CD103+ CD123+ CD11c bright, annexin A1+ , BRAF V600E ; monocytopenia
SBLPN (HCL-v)	CD103+ but CD25– , annexin A1–, BRAF wild-type, MAP2K1; leukocytosis, prominent nucleoli
Splenic marginal zone	Polar (villous) projections , CD103 usually –, CD25 variable, annexin A1–; NOTCH2 / KLF2 ; may have a small IgM paraprotein
Splenic diffuse red pulp	Villous cells, red-pulp pattern, CD25–, annexin A1–; CCND3 mutations; BRAF wild-type
CLL · MCL	CD5+ — CD23+ (CLL) or cyclin D1+ (MCL) — HCL and its look-alikes are CD5–

Annexin A1 + BRAF V600E together settle classic HCL; everything else in this list is negative for both.

PEARLS

- **Splenomegaly + monocytopenia + dry tap = HCL** until proven otherwise
- **BRAF V600E is the diagnosis and the target**. Wild-type → think SBLPN or IGHV4-34
- **Cure-like, not cure**: relapse keeps happening late but responds to retreatment
- **Moxetumomab is gone** (2023) — don't list it as an option